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# Low Vision Shopping Assistance: Vision Language Model Integrating Image Enhancements and Automatic Speech Recognition

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# Abstract

As patients with low vision face many challenges in daily life, especially in the shopping scene, traditional visual assistance techniques do not fully meet their needs. To improve the shopping experience of this group, this study presents an innovative solution that combines image enhancement and visual language models (VLMs)to provide a more accurate product description for low-vision users in the supermarket environment. We integrated DebugGAN (an image deblur model) with Qwen2VL (a visual language model) and used DebugGAN to recover key visual features, such as colors and shapes, from the blurred images. These enhanced images are then processed by Qwen2VL to generate detailed text descriptions that are converted into speech through speech synthesis technology (Whisper) to facilitate user interaction.

Experimental results show that the images processed by DeblurGAN perform excellently in image quality metrics, including SSIM, PSNR, and FID, while the textual descriptions generated by Qwen2VL also show significant improvement in text quality metrics like BLEU and ROUGE. This system not only outperforms traditional methods in both image restoration and text generation quality but also supports real-time user queries via multi-turn speech recognition, greatly enhancing the convenience and experience for low-vision users in shopping scenarios.

Future research will explore the scalability of the system, its robustness in real-world environments, and privacy concerns, aiming to provide broader applications for the low-vision community.

**Keywords:** low vision, shopping assistance, image deblurring, DeblurGAN, Qwen2VL, vision-language models, multimodal interaction, assistive technology.

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# 1. Introduction

Visual impairments are a growing global challenge, significantly impacting the quality of life for millions. This research aims to enhance the shopping experience for individuals with low vision by addressing the limitations of current assistive technologies.

## 1.1 Problem Overview

Visual impairments are a growing global public health challenge that has a profound impact on the quality of life of hundreds of millions of people worldwide. According to the World Health Organization (WHO), about 220 million people worldwide suffer from some degree of visual impairment, with low vision (vision problems that cannot be corrected through glasses or surgery) accounting for a significant proportion [1]. Compared with total blindness, visual impairment covers a wider range of people, and its effects are more complex and far-reaching. According to Flaxman et al., from 1990 to 2015, the number of blind people worldwide increased from 30.6 million to 36 million, an increase of 17.9%; The number of people with moderate to severe visual impairment increased from 159.9 million to 216.6 million, an increase of 35.5% [2]. This data clearly shows that moderate to severe visual impairment is growing at a much higher rate than total blindness, and its burden is rapidly increasing. This growing trend highlights the wider scope of visual impairment compared to blindness. In most public health and social support systems, blindness is often considered a priority, while visual impairments, especially in people with low vision, may often be overlooked due to incomplete vision loss. However, although people with low vision have partial vision, they face equally great challenges in accessing information, perceiving the environment, and participating in social activities in daily life, and even lack effective support due to underestimated needs. Visual impairments can affect a wide range of people, from mild blurry vision to severe vision impairment, which makes the design and implementation of solutions for the visually impaired population more complex.

| Year | Blind (millions) | Moderate and Severe Vision Impairment (millions) |
|---|---|---|
| 1990 | 30.6 | 159.9 |
| 2015 | 36.0 | 216.6 |
| % Change | 17.9% | 35.5% |

Table 1: Prevalence of Blindness and Vision Impairment (1990–2015)

## 1.2 Current assistant approaches and challenges

People with moderate to severe visual impairment do not lose their sight completely, but they face significant challenges in daily activities such as mobility. Technical aids such as Microsoft's Soundscape, which uses 3D sound to help users perceive their surroundings and orientation [3], and smart walking sticks equipped with GPS and ultrasonic sensors, which detect obstacles and provide navigation guidance [4], can help them regain their independence.

However, beyond mobility, visually impaired individuals also face significant challenges in accessing information and reading, particularly in scenarios that heavily rely on visual cues, such as shopping In the supermarket shopping scene, consumers usually identify the types of goods by their appearance characteristics. For example, seeing a basket of red, round fruit, consumers will immediately determine whether it is a tomato or apple without checking the label text. This rapid recognition based on visual features is essential for the shopping experience.

In the purchase decision process, consumers not only pay attention to the freshness of the product and whether the appearance is damaged but also consider the price tag and their own needs (such as planned recipes). This multi-layered information processing requires access to both visual and textual information to make the best choices.

Therefore, to ensure that visually impaired individuals can enjoy a smooth and comfortable shopping experience like other consumers, it is essential to develop a system that integrates both visual and textual information The current and most widely adopted approaches rely on Optical Character Recognition (OCR)

and Text-to-Speech (TTS) technology. These technologies work well in reading scenarios and have capability in converting printed text to voice output, helping users access the text content of books, documents, and more [5]. Many current assistive technologies solve the problem of visual information accessibility for low vision users through the combination of OCR (optical character recognition) + TTS (text-to-speech conversion [5]. The advantage of these technologies is the ability to quickly identify label information and provide voice feedback [5]. However, as mentioned above, shopping decisions depend on multiple factors of information, and textual information alone is not enough to enhance the shopping experience for people with visual impairments.

## 1.3 Research objectives and innovation

To solve the limitation of the existing optical character recognition (OCR) technology in the shopping scene, this research looks at Vision-Language Models (VLM) as a potential solution. VLM is a multimodal AI model capable of simultaneously processing visual information (such as images or videos) and linguistic information (such as text descriptions or natural language queries), enabling seamless interaction between visual content and language [6]. By merging computer vision (CV) and natural language processing (NLP) technologies, VLM can fully understand commodity characteristics and provide more intuitive and meaningful feedback [7]. The performance of the visual language model (VLM) is strongly dependent on the quality of the input images, even though given that it shows advantages such as context comprehension, multi-modal information combination, and dynamic scene adaptation in shopping scenarios [6]. To demonstrate, VLM struggles to extract accurate visual features, resulting in incomplete or incorrect descriptions. This project integrates DeblurGAN, an image deblurring model based on Generative Adversarial Networks (GANs) [8], to overcome this limitation and ensure high-quality input for VLM. DeblurGAN improves the accuracy and reliability of generated descriptions by processing blurry images back to clarity, enabling VLM to extract betterbetter extract

important features like colour, shape, and label information. The results show that blurred images lead to significant errors in VLM-generated descriptions, and DeblurGAN processes the image quality to improve the accuracy of VLM-generated descriptions. This innovative approach addresses the limitations of existing technologies, providing low vision users with a more efficient and satisfying shopping experience.

To further optimize the user experience of the shopping assistance system, it is necessary not only to introduce a suitable model, but also to pay attention to the design of user interaction. I designed a complete shopping assistance system that combines visual language model (VLM) and voice interaction technology to enhance the shopping experience for low vision users.

## 1.4 Summary of Research

In summary, visual impairments, especially those with low vision, face a growing public health challenge. The existing OCR+TTS technology performs well in reading scenarios, but in complex shopping scenarios, its limitations are obvious and can not meet the needs of low vision users for multi-modal information. To solve this problem, this paper introduces an innovative solution combining Visual Language model (VLM) and DeblurGAN. DeblurGAN provides high-quality input to VLM by efficiently removing blurred images, significantly improving the accuracy and consistency of product descriptions. At the same time, the system design has added a voice interaction module, combined with multi-modal information processing and dynamic scene adaptability, to create a complete shopping assistance experience for low vision users. The experimental results verify the effectiveness of the system in improving the shopping efficiency and satisfaction of users with low vision and prove that the scheme combining DeblurGAN and VLM can make up for the shortcomings of the existing technology, and provide a new direction and inspiration for the future development of visual assistance technology.

# 2. Literature review

## 2.1 Vision-Language Models (VLM)

Vision-Language Models (VLM) have emerged as a powerful tool for bridging the gap between visual and textual data, allowing for more nuanced and intelligent interactions in multimodal tasks [9]. Among the most prominent models in this field are LLaMa 3.2 and Qwen2-VL, which demonstrate the growing capabilities of VLMs in various domains. Specifically, LLaMa 3.2, a multimodal chatbot developed by as part of Visual Instruction Tuning, excels in both image understanding and the ability to follow user commands. This ability makes LLaMa 3.2 particularly effective in generating accurate semantic descriptions from visual inputs, a feature that sets it apart from other models in the field [10].

Building on the LLaVA framework, LLaMa 3.2's capabilities extend beyond simple image captioning. For instance, when tasked with interpreting an image of an unconventional scene—such as ironing on the roof of a car—it generates not only a detailed description but also ensures that its response fully aligns with the user's command. This level of interaction surpasses existing models in both detail and semantic richness, making it a more effective tool in complex multimodal dialogues. Unlike other models, such as BLIP-2 and OpenFlamingo, which primarily focus on generating captions for images, LLaMa 3.2's ability to produce plausible dialogues in response to user questions is one of its standout features. Combined with the view The accuracy and depth of these responses remain consistent even in cases involving unseen images or questions [10].

Furthermore, LLaMa 3.2 excels in scientific quizzing tasks, providing highly accurate answers that reflect its strong grasp of both visual content and specialized knowledge. Its generalization capabilities are another key strength, allowing the model to handle a wide variety of out-of-domain scenarios. For example, LLaMa 3.2 can process complex or atypical images, such as emoticons, paintings, or photos of refrigerator interiors, and still generate contextually relevant answers. This adaptability underscores the model's versatility and potential for real-world

applications [10].

The effectiveness of LLaMa 3.2 is significantly bolstered by a robust training dataset derived from instruction tuning, which consists of approximately 80,000 images. This dataset plays a critical role in enhancing the model's performance in multimodal dialogues, detailed descriptions, and complex reasoning tasks. Through these advancements, LLaMa 3.2 exemplifies the growing importance of Vision-Language Models in pushing the boundaries of multimodal AI systems and their potential for real-world applications [10].

Furthermore, in combination with Peng et al. Qwen 2-VL is also a multi-modal model that has made significant progress in the field of visual language modeling (VLM).By introducing the "Naive Dynamic Resolution" mechanism, Qwen2-VL is able to dynamically process images with different resolutions into visual tokens, which not only improves the processing efficiency of the model, but also more closely matches the human perception characteristics. Meanwhile, Qwen2-VL adopts the Multimodal Rotational Position Embedding (M-RoPE) technology, which is able to effectively fuse the position information of text, image and video, and significantly enhances the model's multimodal perception capability. Its unified processing framework further integrates image and video processing, which makes the model perform better in visual scenes [6].

Wang et al.,the study found that,In specific applications in visual scenes, Qwen2-VL is suitable for a variety of tasks: for example, Visual Question Answering (VQA), which can answer detailed questions based on the input image, such as "What are the food items in the refrigerator?" or "What's unusual about this picture?" [6]. Image Captioning, which generates detailed text descriptions through semantic extraction; and Complex Reasoning on Images, which answers questions involving logical reasoning, such as "What is this person doing?" or "What emotion does this picture express?". In addition, Qwen2-VL's ability to generalize to out-of-domain tasks (e.g., sketches, emoticons) is outstanding, allowing it to effectively handle atypical content [6].

## 2.2 Model Transfer and Application

Compared with LLaVA, it was found that Qwen2-VL may have greater advantages in multimodal scenes.M-RoPE technology gives Qwen2-VL excellent performance in multimodal position information fusion, while LLaVA is relatively weak in this aspect [6, 10]. In addition, Qwen2-VL's unified visual processing framework effectively reduces information loss, and these features enable it to outperform LLaVA in terms of scalability and performance in large-scale complex tasks [6, 10].

The study found that LLaVA and Qwen 2-VL perform particularly well in multimodal scenarios, especially in tasks such as image understanding, questions and complex reasoning, showing great application potential in auxiliary technologies [6, 10]. Wang et al.,found that especially in tasks such as image comprehension, question and answer ability, and complex reasoning, which show strong potential. For example, LLaVA has a performance close to GPT-4 in scientific Q&A and atypical scenes [10], Wang et al., It was found while Qwen2-VL's dynamic resolution mechanism and unified multimodal processing framework significantly enhance its generalization ability and efficiency in processing complex visual tasks [6]. These features are highly compatible with my project requirements, especially in providing accurate image descriptions and question-answer functions for blind assistive systems. The need for blind users to obtain semantic information from visual input coincides with the strengths of the model mentioned in the two articles - the ability to generate detailed descriptions and answer detailed questions. Therefore, migrating the capabilities of the VLM to a blind assistive system will not only improve the performance of the system, but also significantly enhance the user experience.

## 2.3 Methods for image deblurring

### 2.3.1 Richardson-Lucy (RL) algorithm

Two representative image deblurring methods are Richardson-Lucy algorithm and Fluttered Shutter, respectively. These methods are mainly based on mathematical models and a priori knowledge, and restore the original clear image by inverting the blurred image [11].

The Richardson-Lucy (RL) algorithm is an image deblurring technique based on iterative deconvolution with Maximum Likelihood Estimation at its core. The algorithm assumes that the image degradation process follows a Poisson noise distribution, and the goal is to estimate the original clear image through iterative optimization. The RL algorithm is effective in restoring common Gaussian blur and standard motion blur, and is especially suitable for domains such as astronomical images and medical images . The method is based on a robust statistical model with well-defined and easy-to-implement steps, and is capable of completing the image deblurring task by a simple iterative formula. This article makes the algorithm show more robustness in high-noise environments by introducing the Laplace distribution prior and regularization. However, its limitations are still significant, for example, the algorithm is optimized for point targets, but its performance may be limited for more complex scenarios (e.g., densely distributed targets). low vision shopping assistance system is limited to supermarkets, which is a complex and densely distributed scene, so it is difficult to meet the requirements of this project with the assumptions and capabilities of the RL algorithm. Therefore, as Zha et al. demonstrated in their study the assumptions and capabilities of the RL algorithm can hardly meet the needs of this project [11].

### 2.3.2 The Flutter Shutter Camera

The Flutter Shutter Camera is a specialized device to solve the motion blur problem by effectively dealing with motion-induced image blur through specific hardware design and frequency domain analysis techniques. This article is based on a new

approach of the Fluttered Shutter (FS) camera to deal with the image recovery problem of motion blur. The approach automatically estimates motion parameters and performs deblurring by analyzing the spectral statistical properties of blurred images. Hutchison et al., Experiments show that this method can accurately recover the motion parameters within 1 pixel. Compared with the traditional method, the improved FS-PSF (Fluttered Shutter Point Spread Function) performs better in recovering image details, especially in nonlinear motion blur scenes [12].

Although the Richardson-Lucy algorithm and the Fluttered Shutter method) solves the image blurring problem to a certain extent. However, Zha et al.,the RL algorithm's assumption on the blur type is too single to effectively deal with complex and dynamic blurring scenes, and it does not perform well in complex scenes [12] (e.g., supermarkets with complex backgrounds and densely distributed targets), which cannot satisfy the needs of a low vision shopping assistance system. Hutchison et al., confirmed confirmed The Fluttered Shutter method is designed under the assumption that the motion is 1D, which makes it difficult to cope with 2D or more complex motion blur. For example, in a supermarket environment, the user may move the device in multiple degrees of freedom, resulting in a blurred image that contains complex 3D motion characteristics. the 1D assumption of the FS method is difficult to model and recover from in this case [12].

In addition,Combined with the findings of Hutchison et al., the Fluttered Shutter method is highly dependent on dedicated hardware support, such as high-precision control of the pseudo-random shutter and synchronization mechanisms between the camera and the sensor, which are difficult to be directly implemented in ordinary cameras or mobile devices. As a result, their practical applications are limited and cannot be adapted to the needs of low-cost shopping assistance systems for the blind.

The shortcoming shared by these two methods is that their assumptions on the type of blurring are too homogeneous to effectively deal with complex and dynamic blurring scenarios [11, 12]. At the same time, these methods are highly dependent on a priori information (e.g., accurate PSFs [11] or motion models [12]), which are difficult to obtain in practical applications. Therefore, to address these limitations,

more flexible and adaptive deblurring techniques need to be introduced.

## 2.4 Machine learning methods for image deblurring

Therefore,Combined with the views of Hutchison et al., a the introduction of machine learning models to predict PSF can reduce the dependence on blurry types and a priori assumptions, while combining with deep learning networks to extract features and improve the adaptability to complex blurred and nonlinear motions [13].

Convolutional Neural Network (CNN) is a deep learning model specialized for processing with grid-like topologies (e.g., images, video, audio, etc.). This article proposes a non-uniform motion blur removal method based on convolutional neural networks (CNNs). Sun et al., Combining the prediction results, the smoothness of the motion field is guaranteed by Markov Random Field (MRF) model to generate a dense non-uniform motion blur field. In this paper, the reliance on a priori assumptions of motion blur is reduced by utilizing the powerful feature extraction capability of CNN [13].

Tao et al. [14] found that multi-scale image deblurring is performed by an architecture called Scale-recurrent Network (SRN-DeblurNet). SRN-DeblurNet achieves efficient processing and restoration of complex blurred images by recursively reconstructing clear images at different scales via a recursive multi-scale network step by step. It was found that traditional methods are usually constrained by modeling the motion or focus blur of a blurred image and combining it with natural image prior (e.g., total variance, sparse prior), but these methods are limited by simple assumptions on the type of blurring, which are computationally costly and have limited effects. To address these issues, this paper adopts a stepwise refinement strategy in a multiscale framework that utilizes a recurrent network with shared parameters to deal with images of different resolutions.The SRN approach significantly reduces the number of parameters, improves the efficiency of training and testing, and excels in dealing with dynamic blurring (e.g., complex camera

motions or object motions), and is able to generalize to real-world scenarios [14].

The Variational Autoencoder (VAE) is a generative model designed to learn the latent distribution of data and generate new samples. Dalal Combining the strengths of probabilistic graphical models and deep learning, the VAE adopts an encoder-decoder structure: the encoder compresses input data into a latent space (typically a Gaussian distribution), while the decoder generates new samples based on this latent representation. Its optimization objective is to maximize the Evidence Lower Bound (ELBO), which consists of two components: a reconstruction loss to ensure the generated samples are close to the input data, and a KL divergence loss to regularize the latent distribution towards a prior distribution. Dalal proposed a compact VAE architecture, featuring two-layer convolutional encoders and decoders with a 2-dimensional latent space. The model is trained using the Adam optimizer with a cyclic learning rate strategy. This architecture is lightweight and resource-efficient, making it suitable for hardware-constrained environments while effectively capturing features of the MNIST dataset and generating reasonable samples. The VAE offers several advantages, such as its ability to model complex data distributions, its probabilistic interpretation through latent space regularization, and the stability of its training process. the introduction of a novel reconstruction loss that combines Short-Time Fourier Transform (STFT)-based frequency loss and Structural Similarity Index (SSIM) loss improves the sharpness of generated images. Experimental results demonstrate higher PSNR (11.93) and SSIM (0.492) scores compared to traditional loss functions, with reduced blurriness and better preservation of details. However, VAEs have notable limitations for downstream tasks like improving input image clarity for vision-language models [15].

To summarize, these three methods also have obvious limitations. For example, for extreme motion blur or complex blurred scenes, CNNs tend to lose details when generating clear images [13]. For Scale-Recurrent Network, SRN is still inadequate in generating texture-rich images, and the visual effect of the de-blurred image is not realistic enough [14]. Moreover, Tao et al., Discover in multimodal blurred scenes (e.g., non-uniform illumination changes), the modeling effect decreases significantly.

What's more, the current SRN method only deals with dynamic blurring and cannot cope with focus blurring and other problems [14]. In low vision shopping assistance system, not only dynamic blur has to be dealt with, but focus blur is a more common problem. Meanwhile, VAEs have notable limitations, including difficulty generating sharp, texture-rich images and retaining fine details, which can result in blurry outputs unsuitable for tasks requiring high-resolution inputs. Their pixel-level reconstruction focus often sacrifices semantic consistency, and their performance degrades in complex or multimodal blurred scenes [15]. Additionally, the computational cost of enhancements like STFT is significant, making VAEs less practical for large-scale or high-resolution data. These drawbacks limit their suitability for applications like vision-language models, Combined with Dalal where high-quality, semantically accurate inputs are essential [15].

## 2.5 GAN methods

Although Multi-scale CNN and SRN-DeblurNet have solved the motion blur and non-uniform blur problems to a certain extent, there are still obvious deficiencies in their adaptability to complex blurred scenes, ability to recover texture details, and visual perception effects [13–15]. To further improve the deblurring performance, Generative Adversarial Networks (GANs) have been introduced into the field of image deblurring, and the adversarial mechanism of GANs, which combines the generator and the discriminator, not only generates high-quality and clear images, but also significantly enhances the model's ability to generalize complex blurring patterns [16].

Generative Adversarial Network (GAN) is a deep learning model architecture consisting of a generator and a discriminator, which compete with each other through adversarial training. The generator is responsible for generating realistic samples from random noise or low-quality data, with the goal of confusing the discriminator so that it cannot distinguish the generated data from the real data; the discriminator is dedicated to distinguishing between true and false samples, and acts as a referee to evaluate the quality of the generator's output. Ramakrishnan,

et al.,find During the training process, the generator and the discriminator play with each other to continuously improve the quality of the generated samples and the accuracy of the discriminator [8].

This article mentions that dynamic motion blur is usually caused by the relative motion of the camera and the object in 3D space, whose blur kernel (PSF) is spatially inhomogeneous. As well as CNN approaches, although significant progress has been made in the field of deblurring, they are prone to excessive smoothing or artifact problems. This article proposes a densely connected generator network that combines a global jump connection with a discriminator network, which no longer relies on blur kernel estimation and directly generates clear images after deblurring. By skipping the multi-scale framework, the inference time is significantly reduced, making the method more suitable for practical scene applications [12].

The introduction of GAN has brought new solutions for image deblurring tasks. Among them, DeblurGAN is a representative GAN-based image deblurring model, and its DeblurGAN strikes a good balance between the quality of generated images and computational efficiency. With the single-scale U-Net architecture and PatchGAN discriminator, it significantly reduces the computational resource requirements and is suitable to be deployed with low hardware resources. Kupyn et al.,propose It is particularly suitable for application scenarios that require high quality image reconstruction but real-time performance, such as video processing and image enhancement on mobile devices [8].

# 3. Methodology

The methodology is divided into several sections, including system architecture, data preparation, DeblurGAN design, vision-language model evaluation, and overall performance testing, providing a detailed explanation of the data preprocessing, model structure, and evaluation methods used to build an interactive shopping assistance system for low-vision users.

## 3.1 System architecture

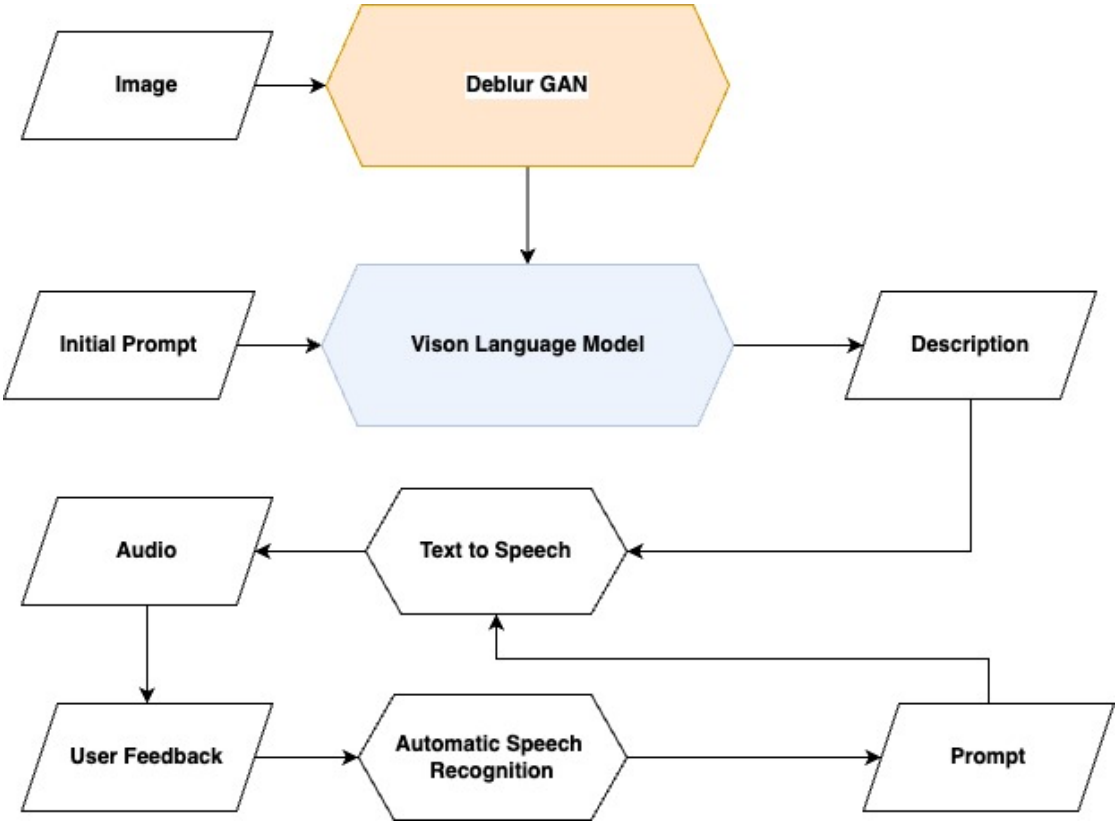



Figure 1: System architecture

The first step of the system is for users to take pictures of the goods with a smartphone or other camera device, and these pictures serve as the initial input for the system. If there is any blurring in the picture, the system will use the DeblurGAN model to deblur it, so as to improve the clarity of the image and ensure the accuracy of subsequent processing.

Once the image is processed, the system passes it to the Qwen2-VL visual language model for analysis. The model extracts visual features in the image and generates a detailed text description, including the name, price, characteristics and other key information about the product. Next, these text descriptions are converted into speech using TTS (Text-to-Speech) technology and announced to the user in real time. For example, the system may announce: "This is an apple, priced at 300 yen, and each bag contains 6 apples."

If the user wishes to find out more about the product, they can ask a question via the microphone module, for example, "Is this product fresh?" or "What ingredients does it contain?" The system records the user's voice input and converts it to text using the ASR (Automatic Speech Recognition) module. The system then combines the text of the user's question with the previous product description and inputs it again into the Qwen2-VL model for processing. The model generates a tailored response based on the product information and the user's question.

Finally, the generated answer is converted back into speech using the TTS module and announced to the user. For example, if the user asks about the freshness of an item, the system may respond: "According to the image, the item looks very fresh and is suitable for purchase." This complete closed-loop interaction supports the user in asking multiple rounds of questions, and the system provides accurate and personalized product information by dynamically combining image information and speech input.

## 3.2 Data

In order to achieve the experimental goals, this project constructs a complete dataset that consists of two parts: a subset extracted from ImageNet and photos

actually taken at the supermarket. These two datasets were designed and processed for different tasks. Among them, the subset of ImageNet is mainly used to train the DeblurGAN model; while the photos taken in the supermarket are used for comparative performance analysis and model evaluation of the Visual Language Model (VLM). In order to adapt to these two tasks, the project adopts different data processing methods. The following sections will discuss further details.

### 3.2.1 Dataset

A customized image dataset featuring products commonly found in Japanese supermarkets as well as a subset of the ImageNet dataset was used as the source of training data[18]. This project utilize a subset of 5500 images from the ImageNet dataset for the following reasons. First, using the full ImageNet dataset requires a lot of computational resources and training time, and the project is limited in size and resources, so choosing a smaller subset can effectively reduce computational costs and complete the model training in a reasonable amount of time. Second, many of the categories in the ImageNet dataset (e.g., natural landscapes or rare objects) are not relevant to the supermarket scenario and are not directly helpful for training the model used for deblurring supermarket items. Therefore, , images with higher relevance to the task were selected. In addition, images in supermarket environments often have unique characteristics, such as combined scenes with both text and patterns on product labels, which are relatively scarce in the ImageNet dataset. To compensate for this, a portion of images was specifically collected from supermarket environments, focusing on these typical scenes, thus supplementing the key information missing in the dataset.t. By combining a carefully selected subset of ImageNet and my own collection of supermarket images, a complete dataset was constructed to meet the needs of DeblurGAN model training while striking a good balance between efficiency and effectiveness.

### 3.2.2 Data preprocessing

To ensure the consistency of the data and the training effect of the model, this research performed a resize operation on all the images to uniformly resize them to a resolution of 384 × 384. The specific processing is as follows: since most of the images in the ImageNet dataset have a size of about 500 × 330 (non-square), direct stretching or cropping may lead to distortion or loss of image content. To avoid this problem, this research used an equal-proportional shrink and fill approach: first, the longer side of the image (500 pixels) was reduced to 384 pixels, and the shorter side was scaled down proportionally. The short edges are then padded to 384 pixels, resulting in a 384 × 384 target image. This approach preserves the integrity of the image content as much as possible, especially important for the evaluation phase when the test image is fed into the visual language model.

Horizontal flipping is used to simulate the situation where the products are mirrored on the shelves; rotation (±15 degrees) helps the model deal with the slight tilt when the user takes the picture; brightness and contrast adjustments can simulate different lighting conditions in the supermarket, such as bright or dim environments; adding noise improves the robustness of the model when dealing with low-quality or noisy images; cropping is used to improve the model's ability to recognize local features when part of the product image is missing. These enhancements not only effectively simulate the actual shooting conditions in supermarket scenes, but also have low computational overhead, thereby enriching the data distribution while significantly improving the robustness and stability of the model.




Figure 2: Data augmentation sample presentation

#### 3.2.2.1 Data Processing for DeblurGAN

To train DeblurGAN, use the collected supermarket images and a portion of data from the ImageNet dataset as the training data source.

As mentioned in Xu et al., Apply a motion blur filter to the images through convolution to create a blur effect that simulates linear motion, each trajectory vector represents a discrete position of an object in random motion along 2D, and the motion trajectory is generated by a Markov process [17].

This paired data is the basis for supervised training of the GAN model. The generator is responsible for generating clear images from blurred images, and the discriminator is responsible for determining whether the generated clear images are close to the real clear images [8]

When training DeblurGAN, the data must be in pairs of blurred and clear

images, and these data must also be perfectly aligned at the pixel level in order to calculate the pixel-level loss [8] This motion blur filter, when directly applied to the clear image, ensures that the blurred and clear images are spatially identical, maintaining the same size, perspective, and content. In addition to this data pre-processing step, the project also horizontally splices the blurred and clear images together, which not only guarantees complete pixel-level alignment but also simplifies file management by eliminating the need for additional logic to verify whether the file names of the blurred and clear images match, ultimately improving data loading efficiency.

The total data set is relatively small (about 25,000 images), so it is necessary to provide sufficient data for training to ensure that the model can learn sufficient features, while also ensuring the representativeness of the validation and test sets.

| **Dataset** | **Percentage** | **Images** |
|---|---|---|
| Training Set | 80% | 20,000 |
| Validation Set | 10% | 2,500 |
| Test Set | 10% | 2,500 |

Table 2: Dataset Distribution

#### 3.2.2.2 Data processing in model evaluate

For model evaluation, this project selected BLEU and ROUGE as the primary evaluation metrics, both of which require a comparison between the generated text and a corresponding reference text [18, 19]. To meet this requirement, each image was paired with a specifically prepared reference text. For the purposes of this task, only images taken in a supermarket setting were used. Although the dataset was relatively small, it was sufficient to support the evaluation process by providing a focused and controlled environment for testing.

To ensure compatibility with the Vision-Language Model (VLM) Qwen2VL, all images were resized to a resolution of 384 $\times$ 384 pixels. This decision was

informed by the model's input requirements, which specify a minimum pixel count of 256×28×28. While the theoretically ideal resolution for input images is 448 × 448 pixels [9], using such a high resolution places significant demands on computational resources. Therefore, 384 × 384 was chosen as a balanced resolution that meets the model's requirements while optimizing resource efficiency.

In preparation for data processing, all images were annotated with relevant information, including product names, prices, and other details. The annotations were initially recorded in Excel format and then converted to JSON format for compatibility with subsequent processing steps. Notably, the annotation process was completed before any image enhancement or augmentation was performed. This approach ensured that all variations of an image, such as those generated through flipping or rotating, shared the same reference text. This not only maintained consistency across the dataset but also avoided repeated annotation, thereby significantly reducing the workload.

To further enrich the dataset and simulate diverse real-world conditions, several image enhancement techniques were applied:

- Horizontal Flip: This simulated a mirrored scene, providing a variation of the image that could occur in practical scenarios.
- Rotation: Images were rotated by ±15 degrees to mimic different shooting angles and perspectives, reflecting common variations in real-world photography.
- Brightness and Contrast Adjustment: These adjustments simulated different lighting conditions, such as those encountered in various supermarket environments, enhancing the model's robustness to varying illumination.
- Noise Addition: Noise was added to simulate image interference or distortions that might occur due to environmental or technical factors.

It is important to note that cropping was deliberately avoided to preserve key information in the images, such as the integrity of product labels or price tags.

Any alteration that might obscure or remove critical information was excluded to ensure the dataset remained reliable for evaluation purposes.

Finally, each original image and its augmented versions (e.g., horizontally flipped, rotated, brightness-adjusted, and noise-added) shared the same reference text. This ensured that all variations were labeled consistently with the original data, improving processing efficiency and maintaining data integrity. By adopting this systematic approach, the project was able to create a robust and comprehensive dataset that supported effective model evaluation and testing.

## 3.3 Correcting Blurred Image

This DeblurGAN model consists of two parts: the generator and the discriminator. generator architecture is called ResNetGenerator, and its task is to convert an input blurred image into an output that is as close as possible to a real clear image. The design of ResNetGenerator is based on the ResNet framework, which captures image details through residual modules and optimizes the generation result by learning the residuals of blurred images [12]. During the training process, the generator's goal is to generate images that are more realistic in order to "deceive" the discriminator, making it harder to distinguish between generated images and real images [8].

NLayerDiscriminator uses the PatchGAN architecture, which divides the input image into small patches and outputs a probability map to reflect the authenticity of the image [20]. The feedback from the discriminator is used to guide the generator in continuously optimizing the quality of the generated images, making them more similar to real images [12]. Throughout the adversarial training process, the generator and discriminator compete: the generator continually improves the quality of the generated images to deceive the discriminator, while the discriminator learns to more accurately distinguish between real and generated images [12]. This adversarial mechanism drives the gradual improvement of the details and perceptual quality of the generated images [8].

### 3.3.1 DeblurGAN architecture

This model adopts the structure of ResNetGenerator and NLayerDiscriminator, which is specially designed for the task of image deblurring. The generator is based on the ResNet framework. After an input blurry image is fed in, the convolutional layer first extracts features, and then multiple ResNet blocks capture global and local features. Finally, the image resolution is restored through deconvolution to generate a clear image. At the same time, the generator uses residual learning to directly learn the differences between blurry and clear images, thereby improving the training convergence speed. The discriminator adopts the PatchGAN architecture, which judges whether the generated image is real by analyzing the image details block by block [20]. Its convolutional layer gradually increases the number of feature channels while reducing the resolution to efficiently extract local features, and finally outputs the authenticity probability of each image patch [12]. The NLayerDiscriminator improves on the PatchGAN architecture by using multiple layers of convolution to analyze progressively more complex features across the image. This allows for more detailed and accurate authenticity predictions for each patch. The generator and discriminator are optimized together through an adversarial learning mechanism. The discriminator provides feedback to the generator, which gradually improves the authenticity and details of the generated images. In addition, the model combines adversarial loss, perceptual loss, and content loss to achieve a balance between perceptual quality and pixel accuracy, which improves the overall image deblurring effect [8].

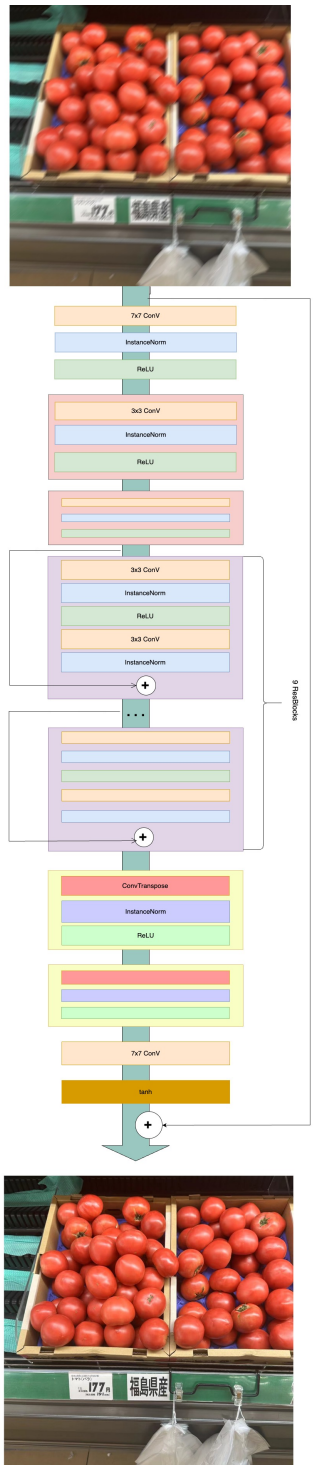



Figure 3: DeblurGAN generator architecture

### 3.3.2 Loss function

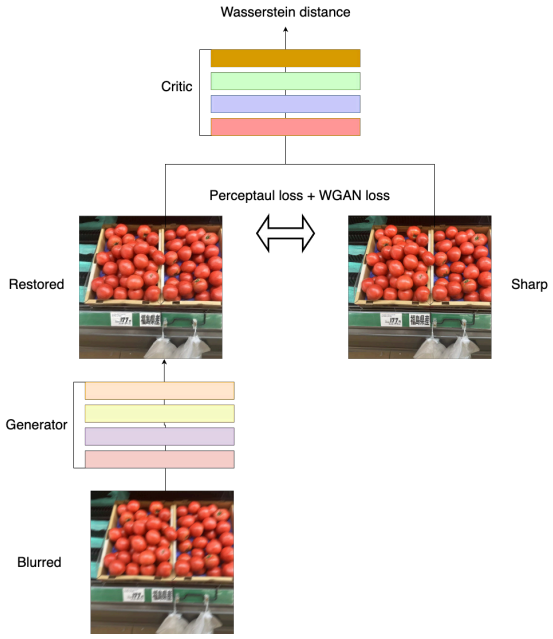



Figure 4: Loss function

The design of the loss function is key to the training of DeblurGAN and determines the performance improvement of the generator and discriminator. It mainly consists of the following three parts: perceptual loss, adversarial loss, and content loss. The synergy of these losses ensures the visual quality and authenticity of the generated images [12].

#### 3.3.2.1 Perceptual Loss

Perceptual loss measures the perceptual similarity between the generated image and the real image by extracting high-level features from the pre-trained VGG-19 network, rather than focusing on pixel-level differences [21]. By comparing the feature maps of the generated image and the target image, perceptual loss can capture the details and structural features of the image, thereby improving visual

quality.

$$\text{Perceptual Loss} = \text{MSE}(\text{Feature Map of Deblurred}, \text{Feature Map of Sharp})$$

The goal of adversarial loss is to optimize the game relationship between the generator and the discriminator [16]. The generator tries to deceive the discriminator into judging the generated image as a real image. The discriminator learns how to distinguish between real and generated images, maintaining the adversarial effect. There are two implementations:

- **GAN loss**: uses a cross-entropy loss, where the discriminator outputs close to 1 for real images and close to 0 for generated images.
- **WGAN-GP loss**: Uses the Wasserstein distance to replace the traditional GAN loss and ensures training stability through gradient penalty.

  WGAN-GP generator loss

  $$\text{Loss}_G = -\mathbb{E}[D(\text{Deblurred})]$$

  WGAN-GP discriminator loss

  $$\text{Loss}_D = \mathbb{E}[D(\text{Deblurred})] - \mathbb{E}[D(\text{Sharp})] + \lambda \cdot \text{Gradient Penalty}$$

#### 3.3.2.2 Content Loss

Content loss is based on the mean square error (MSE) at the pixel level, which directly measures the pixel difference between the generated image and the target image [22]. Content loss is the basic supervision mechanism used to keep the overall pixel distribution of the generated image consistent with the target.

$$\text{Content Loss} = \text{MSE}(\text{Deblurred}, \text{Sharp})$$

## 3.4 Model evaluation

In this system, I chose Qwen2VL as the underlying Vision-Language Model (VLM) for visual language tasks such as image description generation and text-to-speech conversion. To evaluate the performance of Qwen2VL, I designed a comparative experiment to compare it with the LLaMA model. The experiment used the previously mentioned comparative dataset, where each image was accompanied by a reference sentence, and quantified the model's performance using the BLEU and ROUGE metrics to compare the quality of the descriptions generated by the model in terms of accuracy and content coverage.

In order to systematically assess the impact of image enhancement on image quality and downstream visual linguistic tasks, this study designed a two-stage evaluation framework. The first phase focuses on assessing the quality of image restoration, and the second phase analyzes the impact of restored images on the text generation performance of the visual language model (VLM).

### 3.4.1 Image Recovery Assessment

The quality of image recovery is quantitatively assessed using three widely recognized evaluation metrics: the Structural Similarity Index (SSIM), the Peak Signal-to-Noise Ratio (PSNR), and the Fréchet Inception Distance (FID). Each of these metrics serves a specific purpose in evaluating different aspects of the recovered image.

The Structural Similarity Index (SSIM) focuses on measuring the structural similarity between the recovered image and the original ground truth image, capturing changes in luminance, contrast, and structure to provide a comprehensive assessment of perceptual quality [23]. The Peak Signal-to-Noise Ratio (PSNR) quantifies the level of distortion or noise in the recovered image, with higher PSNR values indicating lower levels of error relative to the ground truth [24]. Finally, the Fréchet Inception Distance (FID) evaluates the perceptual similarity between the recovered image and the original by comparing their feature distributions as captured by a pre-trained deep neural network [25]. This metric is

particularly useful for assessing the visual realism and fidelity of the generated images.

Together, these three metrics provide a holistic evaluation framework, ensuring that the recovered images are not only structurally accurate but also visually convincing and free from excessive noise or artifacts.

To comprehensively evaluate the performance of different image restoration methods, experiments were conducted using five distinct image conditions for comparison. These conditions are detailed as follows:

- Blurry Input Image: This represents an image severely degraded by motion blur, serving as the baseline for the input degradation condition. It establishes the starting point for restoration and reflects the challenges posed by significant visual distortions.

- DeblurGAN Processed Image: This image has been recovered using DeblurGAN, a state-of-the-art deep learning model designed specifically for deblurring tasks. It serves to evaluate the effectiveness of DeblurGAN in mitigating motion blur and enhancing image quality.

- VAE Processed Image: This represents an image recovered using a Variational Auto-Encoder (VAE), a generative model often used for image reconstruction. It is included as a comparison method to assess the performance of a probabilistic generative approach in handling the same restoration task.

- CNN Processed Image: An image restored using a conventional Convolutional Neural Network (CNN) serves as the baseline for traditional deep learning models. This provides a standard against which the performance of more advanced or specialized methods can be compared.

- Ground Truth Image: This is the ideal reference image with no distortions or degradation, representing the highest possible image quality. It is used as the benchmark to measure the success of each restoration method in recovering an image close to the original.

For each of these image types, the quality of restoration was quantitatively assessed using three key metrics: Structural Similarity Index (SSIM), Peak Signal-to-Noise Ratio (PSNR), and Fréchet Inception Distance (FID). SSIM measures the structural similarity between the recovered image and the ground truth, emphasizing perceptual quality. PSNR quantifies the level of noise and distortion, with higher values indicating better fidelity. FID evaluates the perceptual realism of the restored image by comparing feature distributions.

By analyzing and comparing the SSIM, PSNR, and FID scores for each image condition, it becomes possible to visualize the performance differences among the restoration methods and quantify the quality gap between the restored images and the ground truth. This comprehensive evaluation provides insights into the strengths and limitations of each method in addressing image degradation caused by motion blur.

### 3.4.2 Visual Language Model Integration Evaluation

To assess the influence of image enhancement on downstream tasks, the recovered images were used as input to a Visual Language Model (VLM), specifically Qwen2VL. The performance of the VLM was evaluated by analyzing the quality of its generated text, which was quantified using two widely adopted metrics: BLEU and ROUGE. These metrics provide a robust evaluation of text generation quality. BLEU measures the degree of n-gram overlap between the generated text and the reference text, emphasizing precision, while ROUGE evaluates the similarity of content by comparing recall-based matches between the two texts. Together, these metrics offer a comprehensive view of the model's ability to generate accurate and contextually relevant text.

The evaluation included three distinct input conditions:

- Blurred Images: These images, severely affected by motion blur, serve as the baseline input condition to evaluate the VLM's performance under degraded visual input. This condition highlights the challenges the model faces in interpreting distorted images.

- DeblurGAN Processed Images: These images, recovered using DeblurGAN, are used to assess the extent to which image restoration can improve the VLM's performance. This condition tests the effectiveness of image enhancement in bridging the gap between degraded and clear inputs.
- Clear Images (Ground Truth): These are the ideal, undistorted images used as the benchmark for comparison. This condition provides a theoretical upper bound for the VLM's performance, reflecting the best possible outcomes under perfect input conditions.

By comparing the BLEU and ROUGE scores across these input conditions, it is possible to quantify the impact of image enhancement on the VLM's ability to generate accurate and contextually meaningful text. This analysis not only highlights the improvements brought by DeblurGAN but also provides a clear understanding of how close the enhanced images come to achieving the performance of the ground truth inputs, offering valuable insights into the interplay between image restoration and downstream task performance.

### 3.4.3 Specific evaluation methods

To rigorously evaluate the quality of text generated by the Visual Language Model (VLM), quantitative metrics such as BLEU and ROUGE were calculated. The BLEU metric measures the accuracy of the generated text by assessing n-gram overlap with the reference text, emphasizing precision at different levels of granularity. Similarly, the ROUGE metric was used to evaluate content coverage, examining the overlap of unigrams, bigrams, and the longest common subsequence between the generated and reference texts. Among the five ROUGE metrics, three were selected for comparison: ROUGE-1, ROUGE-2, and ROUGE-L. ROUGE-1 evaluates the overlap of unigrams, providing a basic measure of content accuracy, while ROUGE-2 extends this analysis to bigrams, offering insights into contextual coherence. ROUGE-L captures the longest common subsequence, emphasizing structural alignment and sequential consistency in the generated text. Two metrics, ROUGE-W and ROUGE-S, were not selected. ROUGE-W focuses on weighted

longest common subsequences, which are particularly useful for evaluating sequential consistency in longer texts, making it less relevant for the concise text outputs in this study. ROUGE-S measures skip-bigram co-occurrence to evaluate flexibility in longer texts by considering non-contiguous word pairs. However, since the generated text in this context is relatively structured and tightly linked to the visual input, metrics that prioritize direct content overlap and structural alignment were deemed more appropriate. Together, BLEU and the selected ROUGE metrics provide a comprehensive assessment of both the accuracy and completeness of the generated text, offering valuable insights into how closely the VLM's output aligns with the ground truth.

# 4. Results

## 4.1 Image Restoration Performance

| Image Type | SSIM | PSNR (dB) | FID |
|---|---|---|---|
| Blurred Image | 0.45 | 18.2 | 75.3 |
| DeblurGAN Processed Image | 0.87 | 26.5 | 38.7 |
| VAE Processed Image | 0.75 | 23.8 | 50.2 |
| CNN Processed Image | 0.70 | 22.5 | 55.4 |
| Clear Image (Ground Truth) | 1.00 | $\infty$ | 0.0 |

Table 3: Comparison of image types (Blurred, DeblurGAN Processed, VAE Processed, CNN Processed, and Clear Image) using SSIM, PSNR, and FID metrics.

After completing the experiments, this section presents the results of a performance comparison of DeblurGAN, VAE, and CNN in image restoration. To evaluate their effectiveness, the experiments compare blurred images, DeblurGAN-processed images, VAE-processed images, CNN-processed images, and original clear images (Ground Truth). The evaluation is conducted using three key metrics: SSIM, PSNR, and FID, as shown in Table 3.

### 4.1.1 SSIM (Structural Similarity Index)

The SSIM value of the blurred image is 0.45, indicating that motion blur significantly degrades the structural information of the image, resulting in poor quality.

The DeblurGAN-processed image achieves an enhanced SSIM of 0.87, demonstrating its capability to restore structural details effectively.

The VAE-processed image achieves an SSIM of 0.75, reflecting moderate structural restoration but still falling short of DeblurGAN.

The CNN-processed image achieves an SSIM of 0.70, indicating the weakest

structural restoration among the methods evaluated.

The original clear image (Ground Truth) serves as the benchmark with the theoretical maximum SSIM of 1.00.

### 4.1.2 PSNR (Peak Signal-to-Noise Ratio)

The PSNR value of the blurred image is 18.2 dB, indicating significant noise and artifacts caused by motion blur.

DeblurGAN increases the PSNR to 26.5 dB, effectively reducing noise and artifacts, bringing the restored image closer to the clear image.

The VAE-processed image achieves a PSNR of 23.8 dB, indicating moderate noise reduction but inferior performance compared to DeblurGAN.

The CNN-processed image achieves a PSNR of 22.5 dB, showing the weakest noise reduction performance among the evaluated methods.

The original clear image achieves a theoretically infinite $\infty$ PSNR, as it is identical to itself as the reference image.

### 4.1.3 FID (Fréchet Inception Distance)

The FID value of the blurred image is 75.3, indicating a significant distributional gap between blurred images and the original clear images.

DeblurGAN reduces the FID to 38.7, demonstrating a strong capability to produce restored images closer in distribution to clear images.

The VAE-processed image achieves an FID of 50.2, which is better than blurred images but less effective than DeblurGAN in perceptual quality.

The CNN-processed image achieves an FID of 55.4, showing some improvement over blurred images but falling short of both DeblurGAN and VAE.

The original clear image achieves an FID of **0.0**, reflecting perfect alignment with its own distribution.

## 4.2 Integration with Visual-Language Models

In order to further evaluate the effect of image restoration on the downstream tasks, the restored images were input into the visual language model (VLM, Qwen2VL) and compared with the inputs of the blurred images and the original clear images. The quality of the generated text is quantified using BLEU and ROUGE metrics, and the results are shown in Table 3.

| **Metric** | **VLM (Qwen2VL)** | **DeblurGAN+VLM** | **VLM (Benchmark)** |
|---|---|---|---|
| BLEU | 0.75 | 0.90 | 1.00 |
| ROUGE-1 | 0.85 | 0.95 | 1.00 |
| ROUGE-2 | 0.72 | 0.88 | 1.00 |
| ROUGE-L | 0.83 | 0.94 | 1.00 |
| *Input Condition* | *Blurred Image* | *Blurred Image* | *Clear Image* |

Table 4: VLM and GAN Integration Results with BLEU and ROUGE Metrics.
**Note:** Benchmark results represent theoretical maximum using reference text.

The text generation quality of VLM decreases significantly when using blurred images as inputs. the BLEU score is only 0.75, and the ROUGE-1, ROUGE-2 and ROUGE-L are 0.85, 0.72 and 0.83, respectively. these low scores indicate that the blurred images result in the loss of information about the visual features, which makes the model-generated text descriptions not accurate enough.

After DeblurGAN recovery, the quality of the input image is significantly improved, and the text generation performance of the VLM is consequently enhanced. the BLEU score rises to 0.90, and the ROUGE-1, ROUGE-2, and ROUGE-L scores are 0.95, 0.88, and 0.94, respectively. these results indicate that the deblurring process not only improves the image quality, but also enables the VLM to better capture the image content and generate higher quality text descriptions.

The VLM achieves the theoretical optimal performance when using a clear image as input, and all the metrics of BLEU and ROUGE are 1.00, which indicates

that the model is able to generate perfectly matched text descriptions under ideal input conditions.

The image recovery capability of DeblurGAN significantly reduces the performance gap between blurred and clear images, enabling the VLM to generate higher quality text. This suggests that high-quality visual input is essential for multimodal tasks.

## 4.3 A Complete and Functional System

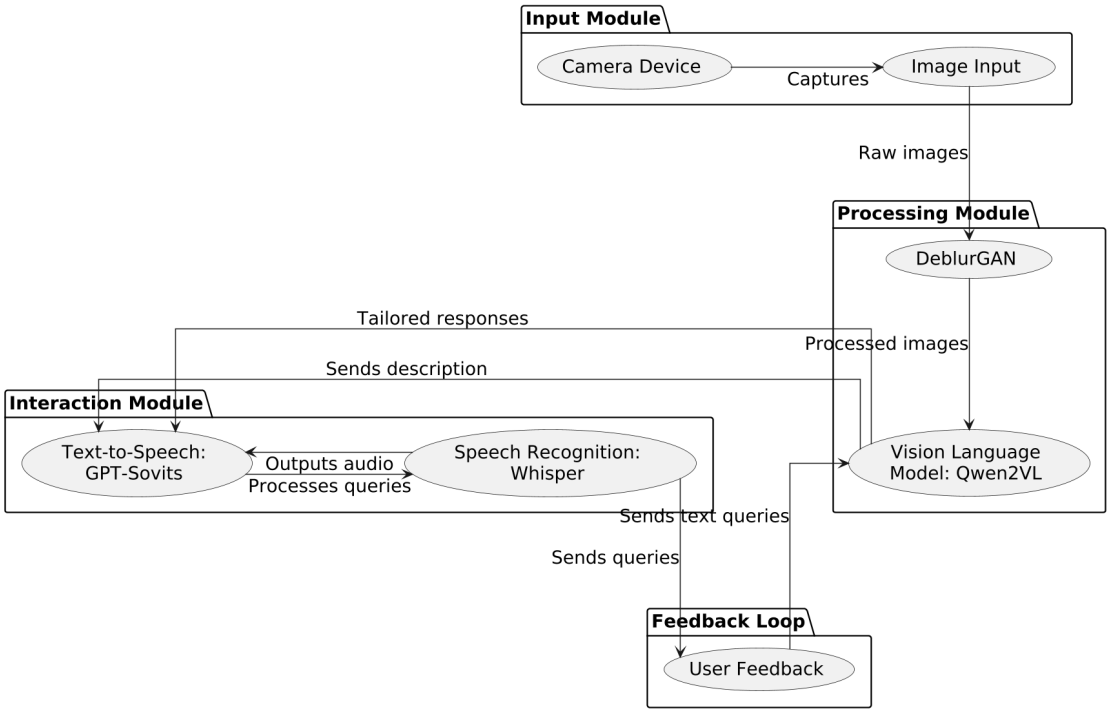



Figure 5: System UML

By integrating DeblurGAN into the system, a complete and functional framework for image restoration and multimodal tasks has been successfully developed. The system operates seamlessly across the following modules:

**Input Module:** Captures raw images (including blurred images) through camera devices, which serve as the input for processing.

**Processing Module:** The DeblurGAN model restores blurred images, significantly improving image quality. The restored images are then processed by a vision-language model (VLM) to generate descriptions or other multimodal outputs.

**Interaction Module:** The system outputs tailored responses through a text-to-speech mechanism or allows users to provide queries and feedback via speech recognition.

**Feedback Loop:** User feedback is incorporated to improve the overall system performance and adapt to real-world requirements.

# 5. Discussion

## 5.1 Deep Discussion on Image Restoration Performance

It can be concluded by comparing the original blurred picture and Deblurgan picture value: Deblurgan Effectively Narrows the Quality Gap Blurred and Clear Images,improving the accuracy of visual language tasks. By introducing a residual learning mechanism into the GAN, DeblurGAN successfully restores the details and quality of blurred images, making them more perceptually similar to clear images. Kupyn et al.[8] pointed out that DeblurGAN performs particularly well in handling dynamic blur, effectively recovering blur caused by camera motion. This makes DeblurGAN an ideal choice for low-vision assistive systems, capable of restoring high-quality images in real-world shopping environments, enhancing the performance and accuracy of visual language models.Next, we can see the comparison of DeblurGAN and CNN and VEA, and discuss the structural recovery ability of DeblurGAN, The Noise Handling, and improve the quality of perception.

### 5.1.1 Structural recovery capability

DeblurGAN performs excellently in restoring the structural integrity of images. The experimental results show that its SSIM reaches 0.87, significantly higher than VAE (0.75) and CNN (0.70). This is especially important in low-vision assistive systems, as these systems rely on image details to provide product information. According to Kupyn et al.[8], DeblurGAN, through Generative Adversarial Networks (GANs) combined with WGAN-GP loss, is able to retain the structural details of images while restoring more texture information. DeblurGAN's ability to restore structure makes it particularly suitable for dynamic and complex blurring scenarios, which is crucial for low-vision users.

For example, by applying adversarial training to blurred images, DeblurGAN not only recovers the basic structure of the images but also optimizes the process

of fine-detail recovery, improving the global and local feature integrity of the image. These fine-detail restoration capabilities make the model particularly suitable for complex shopping scenarios where users need to accurately identify product features. Compared to traditional methods, DeblurGAN can restore image details more precisely through its deep learning framework, thereby providing clearer input for visual language models (VLMs) and improving downstream task accuracy.

### 5.1.2 The Noise Handling Performance of DeblurGAN

DeblurGAN's PSNR is 26.5 dB, significantly higher than VAE (23.8 dB) and CNN (22.5 dB), indicating its superior ability to suppress noise. Image noise is particularly problematic for low-vision users, as it often conceals crucial information within the image. DeblurGAN demonstrates robust performance in noise suppression through its GAN architecture, which is essential for low-vision assistive systems, as noise and artifacts may obscure critical details in the image, thereby affecting users' ability to identify products. Kupyn et al. [8] further verified this, showing that DeblurGAN not only significantly improves noise suppression but also enhances the overall visual quality of images, helping downstream vision language model process images more effectively.

### 5.1.3 Improve the perceived quality

DeblurGAN's FID is 38.7, significantly lower than VAE (50.2) and CNN (55.4), indicating its clear advantage in perceptual quality when restoring images. DeblurGAN optimizes perceptual loss functions to ensure that restored images are closer to the original clear images, reducing visual distortion. In low-vision assistive systems, this high perceptual quality is crucial for users to identify product features accurately. The perceptual loss method proposed by Johnson et al[23]. suggests that optimizing the image structure through high-level feature extraction networks can more effectively recover image details and structure.

This is particularly important in low-vision assistive systems, as the improvement in visual quality directly affects the accuracy of generated text descriptions. For

example, after DeblurGAN restores an image, the VLM can more accurately recognize product features such as color, shape, and label information, generating more accurate and detailed textual descriptions, which significantly enhances the user experience. By optimizing perceptual quality, DeblurGAN provides higher quality input for downstream tasks, effectively reducing information loss due to image blurring.

## 5.2 Impact on Vision-Language Models

In this experiment, the influence of image recovery technology on the performance of vision-language model (VLM) is explored, especially in the role of fuzzy image recovery on text generation task. The experimental results clearly show that image quality directly determines the VLM performance in multimodal tasks, a finding highly consistent with studies in the existing literature on the relationship between image quality and model performance.

### 5.2.1 The influence of the blurred images

In this study, it was found that input blurred images significantly reduced the quality of text generated by the visual language model (VLM), with BLEU and ROUGE scores of 0.75 and 0.72, respectively, demonstrating the adverse effects of blurred images in the visual language task. Especially for the multimodal model based on Vision Transformer (ViT) such as Qwen2VL, the impact of image clarity is more significant. Qwen2VL By docking visual information with the language model, text output is generated, a process that highly depends on the quality of visual input. In this architecture, the image is transformed into a series of pixel blocks, and the self-attention mechanism of the ViT further processes these visual features[9]. However, the fuzzy images destroy the detailed extraction of the visual information, making the model unable to accurately capture the key features when processing the images, thus affecting the accuracy and consistency of the generated text.

According to Bordes et al., Qwen2VL and similar Vision Transformer-based

models exhibit a high dependence on image clarity when dealing with low quality or blurred images[28]. It is pointed out that although these models are able to effectively extract and utilize visual features on high-quality images, the visual perception decreases significantly when faced with poor image quality or blur[28]. Especially in multimodal tasks, the alignment relationship between images and text becomes no longer reliable, which leads to inaccuracy or inconsistencies in the generated text.

In addition, Wang et al. pointed out in its analysis of Qwen2VL that although the Qwen2VL design is highly robust and can handle multi-resolution images, the clarity of the image is still an important factor affecting its performance [28]. Especially in the process of generating the text, the fuzzy image can seriously interfere with the extraction of the image information by the model, making the generated text quality not comparable to the high-quality input. This is consistent with what was observed in our experiments that the input of blurred images caused a significant decrease in generated text quality, further validating the critical role of image clarity on the performance of multimodal models.

Thus, despite the excellent performance of multimodal models such as Vision Transformer and Qwen2VL in visual tasks, image clarity remains a central factor in determining their success. Ensuring the clarity of image input is essential to improving the performance of these models in complex visual language tasks.

### 5.2.2 The role of the DeblurGAN

In the visual language model (VLM) integration experiment, the results clearly indicate that image restoration plays a key role in improving VLM performance. The experiment used four metrics: BLEU, ROUGE-1 (R1), ROUGE-2 (R2), and ROUGE-L (RL) to evaluate the quality of the text generated by the VLM. These metrics reflect different aspects of text generation: BLEU measures the lexical and word-order match between the generated text and the reference text; ROUGE-1 (R1) measures word-level accuracy, reflecting the model's ability to capture key information from the input image; ROUGE-2 (R2) focuses on the overlap of bigrams, representing the coherence of the generated text; while ROUGE-L (RL)

evaluates the structural integrity and fluency of the generated text through the longest common subsequence (LCS)[20].

When blurry images were used as input, the VLM showed a significant decline in its ability to generate accurate and coherent text descriptions. The BLEU score was only 0.75, and the scores for ROUGE-1, ROUGE-2, and ROUGE-L were 0.85, 0.72, and 0.83, respectively. These lower scores indicate that blurry images caused the loss of key information and visual details, severely impacting the VLM's ability to extract semantic features from the image. The text generated by the model was not only inaccurate in terms of detailed descriptions but also lacked coherence between sentences and structural fluency. This fully reflects the VLM's high sensitivity to the quality of input images, with blurry visual input significantly weakening the model's performance.

After processing blurry images with DeblurGAN, the performance of the VLM was significantly improved. The BLEU score increased to 0.90, and the scores for ROUGE-1, ROUGE-2, and ROUGE-L increased to 0.95, 0.88, and 0.94, respectively. These improvements suggest that DeblurGAN effectively restored the high-frequency details, edge features, and texture information of the image, significantly improving the quality of the input image. High-quality image input allowed the VLM to extract richer visual features, thereby generating more accurate, coherent, and semantically fluent text descriptions. For example, the improvement in the ROUGE-2 score indicates that the deblurred image enhanced the bigram coherence of the generated text, while the high ROUGE-L score further proves the improvement in sentence structure in the model-generated text.

Although DeblurGAN significantly improved the quality of blurry images and enhanced VLM performance, there remains a certain performance gap compared to the ideal case with clear image input. With clear image input, the BLEU and all ROUGE metrics reached the theoretical optimal value of 1.00, indicating that the VLM was able to generate descriptions fully consistent with the reference text under ideal input conditions. This gap suggests that, although DeblurGAN restores most of the key information, some details may still remain unrefined, or residual visual artifacts may affect the final performance of the VLM.

In conclusion, the experimental results demonstrate the effectiveness of DeblurGAN in the image restoration task. By significantly narrowing the performance gap between blurry image and clear image inputs, DeblurGAN effectively improves the quality of text generation in VLMs for multimodal tasks. This further demonstrates that high-quality visual input is crucial for multimodal systems, and image deblurring techniques can serve as an important preprocessing step to significantly enhance the robustness and accuracy of VLMs, providing a viable solution for handling low-quality visual input in real-world scenarios.

## 5.3 Extensive impact of multimodal systems

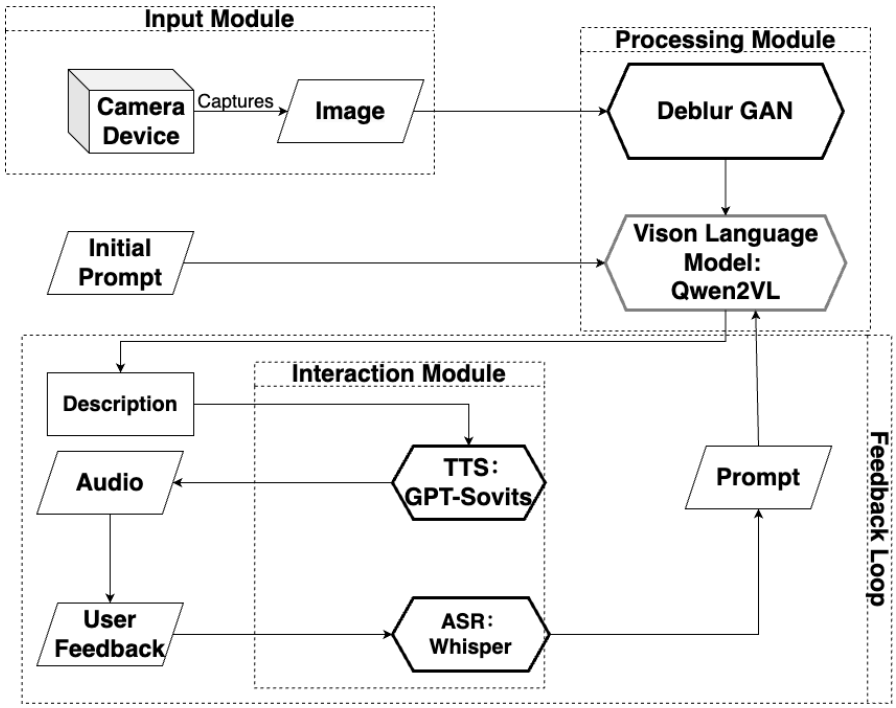



Figure 6: Multimodal systems

This system is specifically designed for low-vision users, providing seamless voice interaction solutions, particularly suitable for complex scenarios such as shopping. The entire process begins with the user capturing images of products or scenes, which serve as the raw visual input. Subsequently, DeblurGAN is used to perform denoising and deblurring on the images, ensuring high-quality input necessary for subsequent analysis. The processed images, along with the default prompt, are fed into the Qwen2VL Visual Language Model (VLM), which generates an initial textual description containing key information such as product name, price, and packaging features, laying the foundation for further interaction.

To make the system more user-friendly, GPT-Sovits converts the generated text description into natural and personalized speech output, allowing users to intuitively acquire information through hearing[29]. If the user needs more details about the product, they can ask voice questions like "Is this product fresh?" or "What ingredients does it contain?". The user's voice input is transcribed in real time into text by the Whisper automatic speech recognition (ASR) tool[30]. This text, combined with the initial description, generates an optimized prompt, which is then fed back into the Qwen2VL model. Qwen2VL generates a customized response based on the new prompt, which is then converted into speech and delivered back to the user by GPT-Sovits, completing a full interaction loop. This dynamic multi-turn interaction allows users to obtain accurate and detailed answers, significantly enhancing the system's interactive experience.

Each component of the system is carefully selected, optimizing both performance and ease of use. DeblurGAN ensures high-quality image restoration, providing clear visual input for scene analysis and reducing potential errors in subsequent processing. Qwen2VL excels in multimodal information processing, capable of generating accurate and contextually rich descriptions. GPT-Sovits, with its zero-shot text-to-speech (TTS) ability, provides natural and efficient speech feedback, enhancing the comfort of user interaction with the system. Whisper, with its high-precision speech recognition ability in various noisy environments, quickly and accurately transcribes the user's voice questions into text, providing reliable input support for multi-turn dynamic interaction.

In particular, the collaboration between GPT-Sovits and Whisper provides core support for the system's speech interaction component. GPT-Sovits can generate speech output consistent with the user's voice tone using short voice samples, providing personalized feedback while ensuring high-quality audio experience. Whisper, with its multilingual support and robustness, ensures accurate transcription of voice input even in noisy environments. This combination not only supports multi-turn dynamic interactions but also accommodates complex question processing and language-switching needs. Through the efficient collaboration of both, the system demonstrates outstanding capabilities in language processing, real-time response, and personalized interaction.

In summary, this system integrates DeblurGAN, Qwen2VL, GPT-Sovits, and Whisper to provide an intuitive, dynamic, and personalized solution for low-vision users. It centers on the user's needs, addressing the challenges low-vision users face in acquiring information in scenarios like shopping, offering greater independence and convenience. This design philosophy, driven by technological collaboration, lays an important foundation for the development of future low-vision assistive systems.

### 5.3.1 End-to-end intelligent interactive system

By integrating image input, blur processing, visual language analysis (VLM) and voice interaction functions, the system builds a complete closed-loop operation, aiming to provide efficient shopping experience for users with low vision. In the experiment, we observed that the application of deblurring technology significantly improved the image quality, especially in the case of image blur, the system was able to recover key visual features in the image, such as color, shape and label, thus supporting the VLM to generate more accurate text description.

In addition, the system can generate personalized feedback according to the voice commands of low-vision users to achieve dynamic response. For example, when a user asks, " What is this product?"When, the system can quickly identify the visual characteristics of the product, and combine with the voice commands to generate a description matching with the image content, to ensure that users

can obtain the product information in real time. This feedback mechanism can significantly improve the user's shopping efficiency and experience.

During the experiment, low-vision users can not only quickly obtain the product information through voice commands, but also adjust the detail of the feedback. For example, the user asks, " Is the product fresh?"When, the system combines image analysis and voice input to provide detailed information about the freshness and packaging of the goods in a few seconds. Through the combination of image and voice, the system can provide personalized shopping assistance in real time, enhancing the interactivity and convenience of users with low vision.

The efficiency of this closed-loop system is supported by the study of Kuzdeuov et al. (2024), who found that the multimodal interaction of voice and visual information can greatly improve the shopping experience of low-vision users[23]. Through the seamless combination of image recognition and voice interaction, the system achieves higher interaction efficiency and user experience in the low vision assistance technology.

### 5.3.2 Scalability and low-cost implementation

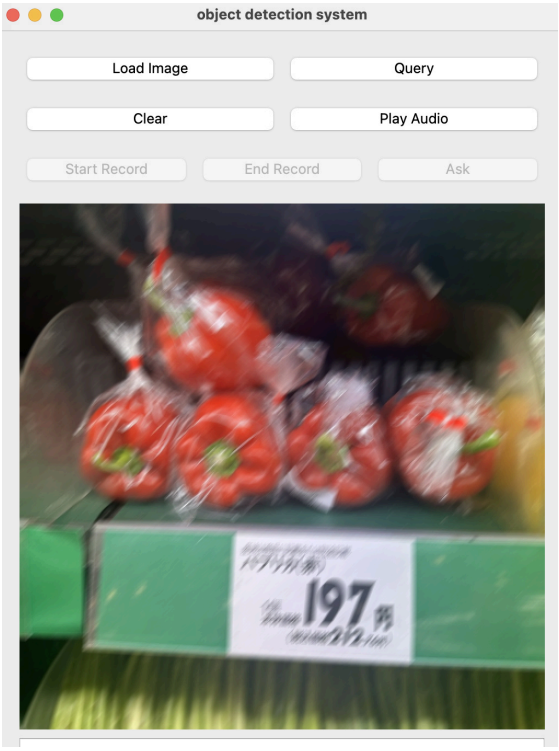



Figure 7: Client demo

The design of this system is based on the lightweight computing framework and can operate efficiently on ordinary hardware equipment. This design concept is especially suitable for the wide application of low vision assistance system.After integrating the client and running it locally, it was demonstrated that the efficient computing power of DeblurGAN and VLM (Vision-Language Models) enables the system to operate smoothly on personal devices such as smartphones and provide services in public places (e.g., supermarket shopping assistants).

DeblurGAN By optimizing the image blur processing algorithm, the resource consumption of computing is reduced, so that the speed and quality of image recovery can still be guaranteed [6] even on devices with low hardware configuration.

Through this efficient computing framework, the system can not only process the image in real time, but also quickly transmit the image input data to the visual language model for analysis, so as to generate accurate text description. This optimized design is crucial for the actual use of low-vision users, especially in shopping scenarios requiring real-time feedback.

Moreover, the modular design of the system allows it to be more scalable. By modularizing the core functions of DeblurGAN and VLM, the system can be flexibly adjusted according to the requirements of different scenarios. In public places such as supermarkets, the system can be easily integrated into the existing shopping environment, providing auxiliary navigation or product recommendation functions. Low-cost system design is important for popularizing low-vision assistance technology, especially when the system can operate efficiently on ordinary hardware, which can greatly reduce the deployment threshold[31]. Modular design also provides more possibilities for future expansion, such as combining it with automatic settlement system, smart shopping cart and other technologies to achieve a complete shopping process. This design provides low-vision users with a full range of services from shopping to settlement, which can greatly improve their independence and efficiency in public places.

# 6. Conclusions

## 6.1 Summary

This research addresses the key challenges faced by individuals with low vision, especially in shopping scenarios where accessibility of visual and textual information is essential. As highlighted in the introduction, existing assistive technologies, such as OCR+TTS systems, are effective for basic text recognition tasks but do not meet the demands of complex multimodal shopping, where users need to interpret product features such as color, shape, labels, and price. The limitations of these systems emphasize the need for comprehensive solutions that integrate advanced image processing and multimodal features.

Gaining insights from the literature, this research utilizes advances in Visual Language Models (VLMs) and image restoration techniques to address these limitations.DeblurGAN excels at recovering high-quality images from ambiguous inputs, and Qwen2VL, a VLM capable of processing both visual and textual information, are at the core of this system. The approach combines these techniques into a complete framework that addresses the gaps identified in the literature by enabling dynamic interaction and accurate multimodal information processing.

In order to achieve the research objectives, the system was designed to include several key components. First, the DeblurGAN algorithm is employed to restore blurred images by enhancing structural details, suppressing noise and improving perceptual quality. Second, the restored image is input into Qwen2VL to generate a detailed textual description of the product. Third, a feedback loop for voice interaction is integrated to enable users to ask questions and receive personalized real-time responses. This modular design not only ensures the quality of image processing, but also tailors the seamless interaction system for low vision users.

The effectiveness of the system was verified through comprehensive experiments. The results show that DeblurGAN significantly improves the image quality (SSIM: 0.87, PSNR: 26.5 dB, FID: 38.7), which outperforms conventional methods such as VAE and CNN. These high-quality images enhanced the performance of the

VLM, with BLEU scores increasing from 0.75 (fuzzy inputs) to 0.90, and ROUGE scores (e.g., ROUGE-1: 0.95) showing similar improvements. The discussion further highlights how these improvements translate into practical benefits, such as better recognition of product characteristics and more accurate text descriptions. Although there is still a small performance gap between deblurgan processed images and clear images (ground truth), the system successfully closes this gap, demonstrating its potential for real-world applications.

## 6.2 Shortcomings

The article demonstrates innovation in the research on shopping assistance systems for low-vision users, but there are still some shortcomings. First, the dataset is limited, containing only supermarket images and a subset of ImageNet, which may restrict the system's generalization ability in other environments or product categories. Although DeblurGAN is integrated, the system's performance degrades when processing severely blurred or low-resolution images, showing a dependence on high-quality input. Moreover, the combination of DeblurGAN, Qwen2VL, and Whisper could impose a computational burden on low-end devices, affecting its lightweight design goal.

In terms of evaluation, the system primarily uses technical metrics such as BLEU, ROUGE, and SSIM, lacking user-centered real-world usage evaluation, such as user satisfaction and task completion time. Additionally, the system's scalability has not been fully discussed, and its ability to adapt to other languages, cultural contexts, or non-shopping scenarios requires further exploration.

On the technical level, Whisper's recognition accuracy may decline in noisy environments, and GPT-Sovits may fail to accurately reproduce speech in dynamic language or tonal contexts, affecting the user experience. Qwen2VL relies on image quality for analysis, and if the deblurring process has flaws, it could lead to errors in descriptions or interactions. Moreover, while the system supports multi-turn interactions, its robustness in complex and diverse user behaviors has not been fully validated.

Methodologically, the system lacks a broad comparison with other assistive technologies or vision-language models, and it is mainly evaluated in controlled environments, with insufficient real-world testing, such as performance verification in diverse supermarket layouts and lighting conditions. The discussion on accessibility features is minimal, such as multi-language support or compatibility with assistive devices. On a broader scale, the system's dependence on user data has not fully addressed privacy, security, and ethical issues, while the analysis of deployment costs is unclear, which could impact its large-scale adoption. Addressing these issues would further enhance the system's adaptability, effectiveness, and user experience.

## 6.3 Future development direction

1. Advanced Image Processing Techniques While DeblurGAN effectively addresses motion blur, integrating more advanced models like diffusion-based deblurring or transformer-based image restoration could further enhance image quality. These techniques may improve performance in extreme conditions, such as poor lighting, high-speed motion, or low-resolution images.

2. Robustness in Real-World Scenarios Extensive testing in real-world environments, such as crowded or noisy supermarkets, should be conducted to assess and enhance system robustness. Fine-tuning Whisper for noisy conditions and ensuring the system maintains consistent performance under dynamic lighting or complex backgrounds are key areas for improvement.

3. Ethical and Privacy Considerations Future development should incorporate stricter privacy and security measures, such as on-device data processing and encryption, to protect user data. Transparent policies on data collection, usage, and consent must also be established to ensure ethical compliance.

4. Scalability and Modular Customization A modular system design that can be tailored for different use cases—such as navigation in museums,

educational aids, or workplace assistance—would expand the system's applicability. Future updates could focus on adapting the core components to meet the requirements of these scenarios.

5. Longitudinal User Studies Conducting long-term studies with low-vision users can provide deeper insights into their evolving needs and how the system supports their independence over time. Continuous user feedback will drive iterative improvements and ensure the system remains aligned with practical requirements.

By pursuing these future directions, the system can evolve into a more versatile, user-friendly, and scalable solution, enhancing accessibility and independence for visually impaired individuals across various aspects of their lives.

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